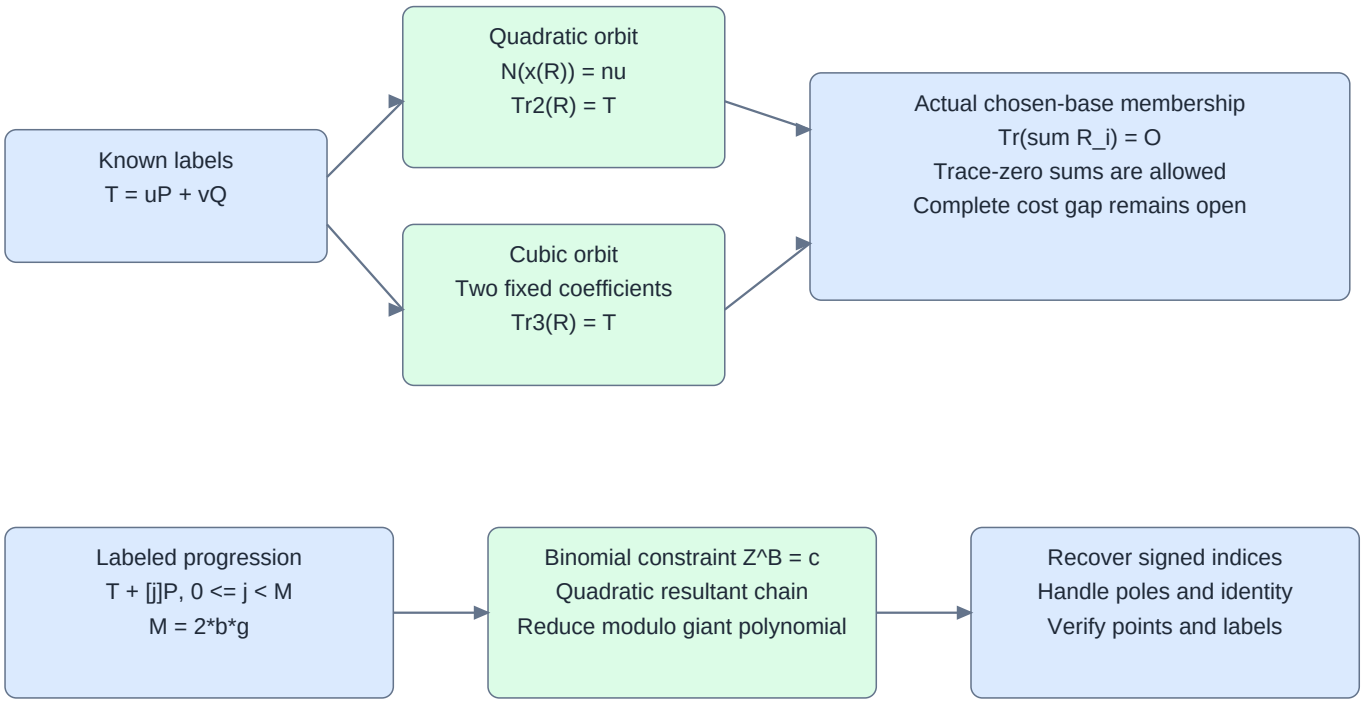
Frobenius lifts and progression certificates

ARCHIVAL REPORT / PRE-GATE. Three prototypes, their original records and written arguments are published for review. No complete ECDLP cost improvement or qualified timing speedup is established.

Quadratic: fixed coordinate norm with known elliptic trace; reciprocal sextic divisors. Cubic: two fixed orbit coefficients; a quartic finds lifts, and formal quotients avoid unnecessary common-field embeddings for trace-only computations.

Progression blocks: preserve the raw binomial through homogeneous quadratic resultants, recover signed indices, and check actual points and labels.



No isogeny edge is asserted. Diagram describes the archived algorithms.

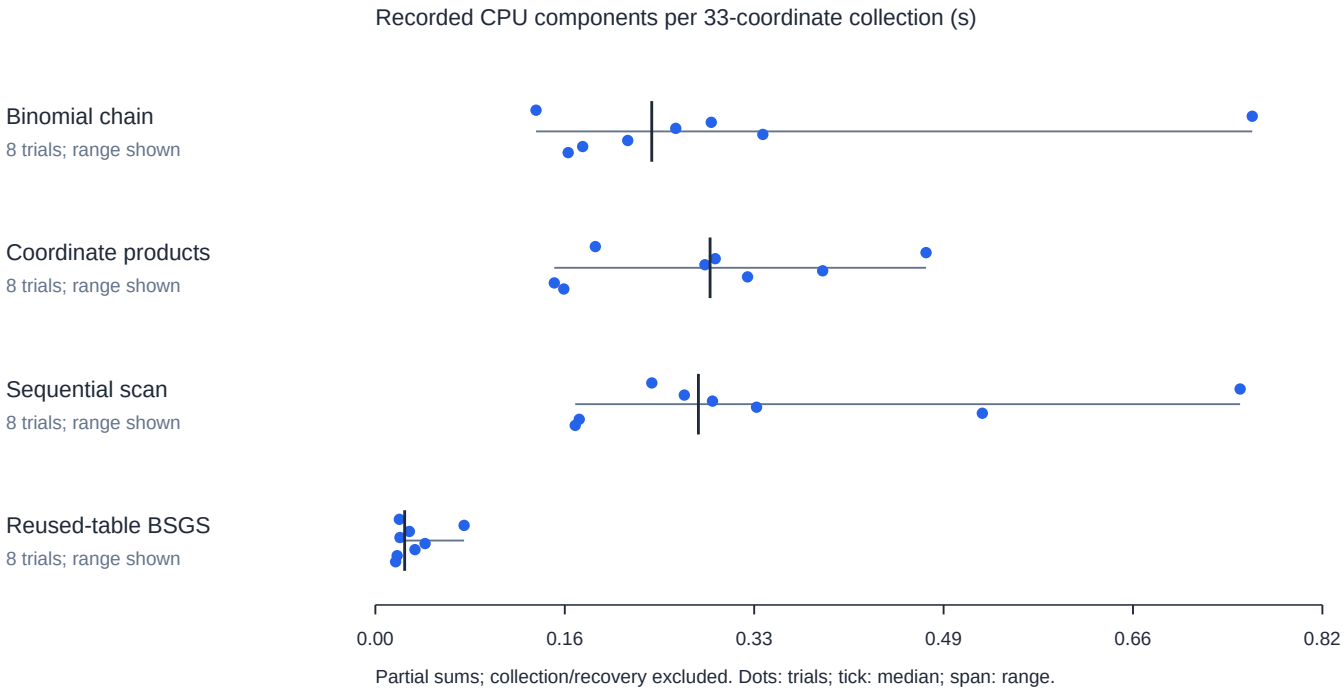
The original combined requirement remains open: a cheap known-label construction into a small usable coordinate domain, together with lower total time to verified relations. Auxiliary trace labels do not give labels for each extension point in the original prime-order group.

The six/nine-point identities retain actual chosen-base membership and the full condition Tr(sum R_i)=O. In the quadratic checks, 302 of 304 conjugate choices had a nonidentity extension sum in the trace-zero kernel.

Sources: archive/WRITEUP.md; archive/experiments/ideas/lift_benchmark.json; archive/experiments/ideas/lift_witnesses.json. No new scientific experiment was run for this report.

Historical partial construction components

Input: y^2=x^3+2*x+25 over F_65537, full prime order 65,371. Each trial collects all 33 usable abscissae of a 64-coordinate coset, using blocks of length 2,048. All methods receive identical starts and stopping rules.



Recorded component sums: eight trials per method

Method	Partial CPU seconds	Counted group operations
Binomial chain	2.351736994	45,973
Coordinate products	2.276435383	45,973
Sequential scan	2.755015272	1,080,978
Reused-table interval BSGS	0.262263897	77,662

CORRECTION: these partial sums omit sorting hits, maintaining known coordinates, termination and transcript work, and antipodal scalar recovery/verification. Each method has 353 antipodal opportunities; the recovery E.mul() calls are uncounted. Complete construction costs and their ratios are unknown. Archive bytes and numbers are unchanged; see ERRATA.md.

Counted: charged setup, sampling, successful/failed solves, solver index recovery, and scalar_verify() hit/identity checks. Per arm: 515 blocks, 58 empty. BSGS reuses a 2,048-point table. CPU-only; isolation/placement and qualified uncertainty receipts are absent. No qualified speedup is assigned.

All trials and sample ranges are shown. Raw source: archive/experiments/ideas/block_benchmark.json. The chain retains M-scale field work; the conjectural O~(B+sqrt(M)) total bound was not achieved.

Small prescribed domains: the scoped restriction

Statement (written proof, not Lean-checked): with fixed nonzero norm and nonrational quadratic abscissae, B prescribed values of the orbit trace coordinate cover at most 2B nonzero elliptic traces. The same bound holds for B prescribed h values in irreducible cubic polynomials Z^3+h*Z^2+d*Z-c, with c,d fixed.

Proof: a prescribed quadratic orbit polynomial determines one conjugate pair of abscissae. At one root, the curve equation has at most two ordinates; Frobenius determines the other ordinate. The resulting traces are T and -T. A prescribed irreducible cubic similarly determines one orbit of three abscissae, with at most two ordinate orbits and opposite traces. Taking a union over B parameters proves the bound.

Therefore independent uniform labeled rejection sampling has nonzero restricted-lift success probability at most 2B/r. This does not rule out nonuniform constructions, adaptive domains, higher-degree chains, or a better direct relation solver. The unrestricted auxiliary norm domain still has about p abscissae.

Checks, provenance and remaining obligations

The stored audit verifies 1,664 known labels, 2,496 extension representations, 78 principal divisors, 588 divisor zeros and 1,163,264 progression indices. The extension verifier uses separate field arithmetic; prime-field group/scalar primitives are shared with the producer. No fully independent implementation of every layer is claimed.

The 34-file archive is preserved byte-for-byte from local commit 820693c998f88b4f7ee213c70c8b99be23734f1b. Its manifest and PROVENANCE.json bind the import. Available original logs are retained. Historical timings are exploratory and are not promoted to current lab validation or candidate records.

Lean formalization remains open; no Lean executable was available on the publication host. A faithful proof must establish the geometric/Frobenius orbit fiber statement, not assume the fiber bound as an axiom. The report keeps the full written statement and scope.

Next implementation obligation: solve the reciprocal coefficient surface while retaining chosen-base membership, or reduce the progression certificate below its present field-work cost. A future measurement must use frozen inputs, full failures/extraction/verification accounting and current lab controls.

References: Petit-Kosters-Messeng, PKC 2016 (rational-map factor bases); Amadori-Pintore-Sala, ePrint 2017/609 (known-label setting). General map families and classical resultant tools are not claimed as new. See README.md and archive/WRITEUP.md for links, formulas and exact limitations.